Master Document 4: Role-Based Access Control (RBAC) & Feature Specifications

Project Name: Agri-Grid: Smart Agriculture & Food Security Ecosystem
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Context: System Security, User Permissions & Dashboard Features

1. The RBAC Security Gateway & Data Isolation

The Agri-Grid ecosystem manages sensitive financial transactions, commercial crop data, and government auditing ledgers. To satisfy strict non-functional security requirements, the platform relies on Role-Based Access Control (RBAC).

- **Authentication & Routing:** Upon login, the security gateway verifies the user's credentials and generates a role-specific session token. This token automatically routes the user to their designated endpoint (e.g., /dashboard/farmer or /dashboard/buyer).
- **Data Isolation (Zero-Trust):** The system enforces strict URL and API endpoint protection. A Transporter attempting to manually navigate to the Farmer's financial ledger will be blocked by the server. Users only possess "Write" and "Read" access to the specific database tables relevant to their operational duties.
- **Admin MFA:** High-level administrative roles (Compliance Officers) must pass Multi-Factor Authentication (MFA) via SMS OTP or an authenticator app before accessing the financial ledgers.

2. The 6 System Actors & UI Feature Specifications

2.1. The Farmer (Supply Side)

Access Level: Full read/write access to personal crop data, listings, and subsidy claims. No access to buyer escrow pools or agronomist queues.

- **Active Farm Overview (CRUD Logs):** A digital ledger where the farmer performs Create, Read, Update, and Delete operations on their active fields, planting dates, and acreage.
- **Predictive Yield Forecaster:** A calculator that triggers the Python backend to ingest API weather/soil data and outputs an estimated harvest date and yield volume for a specific crop.
- **B2B Marketplace Publisher:** Converts the yield forecast into a public marketplace listing (generating a unique Batch ID) where the farmer can set an asking price and available pickup dates.
- **Diagnostic Support Studio:** A ticketing inbox where farmers upload high-resolution photos of diseased crops, select visible symptoms, and receive digital treatment prescriptions from Agronomists.
- **Compliance & Subsidy Ledger:** A portal to apply for sustainability programs (e.g., Drip Irrigation Grants) by uploading proof-of-practice documents for government review.

2.2. The B2B Buyer (Grocer/Restaurant/Wholesaler)

Access Level: Read access to the public crop catalog and personal order tracking. Write access strictly for creating purchase contracts and releasing escrow.

- **The Procurement Exchange:** A robust, filterable catalog of active produce listings (filterable by crop variety, location radius, and organic status), powered by complex SQL SELECT queries.
- **Escrow Checkout & Smart Contracting:** A checkout screen where buyers define destination warehouses and mandate Cold-Chain Parameters (e.g., "Must be kept below 6°C"). Funds are locked in a digital Escrow rather than paid upfront.
- **Live Cold-Chain Transit Tracker:** An interactive map dashboard displaying the GPS progress of their order alongside a live line-graph tracking the interior temperature of the delivery truck.
- **Quality Acceptance & Settlement:** A digital sign-off terminal. Upon physical delivery, the buyer inspects the goods, leaves a rating, and clicks "Accept," triggering the backend to release the escrow funds to the Farmer.

2.3. The Transporter (Logistics Driver)

Access Level: Mobile-optimized read access to assigned dispatch manifests. Write access to update delivery statuses and log telemetry. No access to financial transaction values.

- **The Active Dispatch Manifest:** A mobile-friendly list of assigned freight runs, displaying pickup/drop-off coordinates and temperature constraints, automatically generated when a Buyer initiates a purchase.
- **The Custody Handshake (QR Scanner):** A browser-based camera interface. The driver scans a QR code at the farm to officially transfer liability, changing the database status to "In Transit."
- **The Cold-Chain Telemetry Logger:** A dashboard displaying the truck's current internal temperature. Features "Offline Flash Caching," storing telemetry in the mobile browser if cell service drops, and auto-syncing to the database once reconnected.
- **Proof of Delivery (PoD) Terminal:** A digital signature pad for the Buyer to sign upon arrival, coupled with an image upload button for the driver to prove the cargo was delivered intact.

2.4. The Agronomist (Plant Telemedicine)

Access Level: Read access to farmer diagnostic tickets and specific farm environmental data. Write access to issue prescriptions. Must be manually credentialed by a system admin.

- **The Triage & Ticketing Queue:** A Kanban-style board ("New," "In Review," "Resolved") where the agronomist can filter incoming help requests by crop specialization and "claim" a ticket to prevent duplicated work.
- **The Diagnostic Workbench:** A split-screen UI displaying the farmer's uploaded leaf photos alongside an automated API-driven "Farm Context" sidebar (showing 14-day rainfall and soil moisture for that specific farm).
- **The Digital Prescription Pad:** A structured medical form allowing the expert to select standardized diagnoses, type treatment plans, and push a digital prescription back to the farmer's dashboard.
- **Regional Outbreak Analytics:** A macro-level heatmap plotting resolved disease tickets

by geographical coordinates, allowing the agronomist to issue mass warnings if a regional outbreak (e.g., Locusts or Blight) is detected.

2.5. The Compliance Officer (Government/NGO Admin)

Access Level: Master read-only access to all system activity for auditing. Write access exclusively for approving/rejecting subsidy claims and initiating fund disbursements.

- **The Subsidy Claims Queue:** A secure, sortable table of pending grant applications submitted by farmers, containing uploaded evidence (e.g., water meter photos).
- **The Environmental Verification Engine:** A comparative split-screen that automatically calculates a "Confidence Score" by checking the farmer's manual claims (e.g., reduced water usage) against the independent API historical weather/soil data.
- **The Financial Disbursement Ledger:** A secure portal displaying the remaining NGO/Government budget. Allows the officer to execute ACID-compliant SQL transactions to disburse approved funds into the farmer's wallet.
- **System-Wide Audit Logs:** A sterile, read-only history of every major action taken on the platform (listings created, escrow released, telemetry logged), exportable to CSV/PDF for government compliance records.

2.6. The Public Consumer (End-User)

Access Level: Unauthenticated, entirely read-only access to specific public Batch IDs.

- **QR-Code Provenance Storyboard:** The consumer scans a physical QR code on their groceries. The system queries the database and renders a mobile-friendly visual timeline showing the farm of origin, agronomist health checks, and the transporter's temperature logs, proving the food's safety and authenticity.